

RECENT POSTS

Bluetooth 5.0 OBD2 Scanner for iPhone & Android with J1939 Support and Real-Time Engine Diagnostics

25 Real-World Projects Using the PiCAN-M Raspberry Pi CAN Bus HAT

OBDII Breakout Box Review: The Ultimate Tool for ECU Programming, Protocol Detection, and Automotive Diagnostics

Teensy 4.0 NMEA 2000 Development Board with IPS LCD – The Ultimate Marine Electronics Prototyping Platform

Open-Source Tool Simplifies CANopen Development and Testing

[Home](#)

[Blog](#)

[Raspberry Pi and PiCAN-M in Marine NMEA 2000 Systems: Power Supply Considerations](#)

Raspberry Pi and PiCAN-M in Marine NMEA 2000 Systems: Power Supply Considerations

Posted by Wilfried Voss on Jan 30th 2026



Single-board computers based on the Raspberry Pi have become common building blocks in modern marine electronics. When combined with PiCAN-M, they are frequently used as onboard data gateways for applications such as Signal K and OpenPlotter, translating NMEA 2000 traffic into IP-based data streams, dashboards, and logs.

In practice, most field issues reported in these systems are not related to CAN communication itself, but to **power budgeting and power stability**, especially when the Raspberry Pi is powered directly from the NMEA 2000 backbone.

This article explains how Raspberry Pi power requirements differ across versions 3, 4, and 5, how those requirements interact with the NMEA 2000 power model, and how the PiCAN-M variants with and without onboard SMPS should be used in each case.

NMEA 2000 Power Model in Brief

NMEA 2000 is designed around a shared network power concept. Devices are powered from the backplane using Equivalent Load Numbers (LEN), where 1 LEN corresponds to 50 mA at nominal network voltage.

In real installations, the following assumptions are common:

- Backbone power is nominally 12 VDC (with an allowed operating range).
- Total backbone current is typically limited to about 3 A by cabling, fusing, and installation practice.
- Devices drawing more than approximately 1 A equivalent are normally expected to be powered separately and connected to the network for data.

This model works very well for sensors, displays, gateways, and control units that draw modest and predictable power, but is not well suited to powering general-purpose computers.

Raspberry Pi Power Requirements by Generation

The Raspberry Pi family spans multiple performance classes. While they share a 5 V input requirement, their **transient behavior differ significantly**.

Raspberry Pi 3

Raspberry Pi 3 models are specified for a 5 V supply rated at approximately 2.5 A.

In marine gateway applications (Signal K, OpenPlotter, logging, light dashboards), the Pi 3 typically operates within its power envelope. Peak current events exist, but they are relatively modest.

From a power perspective, the Raspberry Pi 3 is generally compatible with small, well-designed DC/DC converters.

Raspberry Pi 4

Raspberry Pi 4 models increase CPU performance, memory bandwidth, and I/O capability. The recommended power supply is 5 V at 3 A.

In steady operation, a Pi 4 used as a marine gateway often runs below this limit, but transient peaks occur during:

- boot
- CPU bursts
- Wi-Fi activity
- SD card writes

This places the Pi 4 at the **upper edge** of what a 5 V / 3 A rail can reliably support.

Raspberry Pi 5

Raspberry Pi 5 represents a major architectural jump. It is specified for a 5 V supply rated up to 5 A.

Even when no USB SSDs or high-power peripherals are attached, the Pi 5 can draw substantially more due to higher CPU performance, I/O bandwidth, and power management behavior.

A 5 V / 3 A supply is sufficient only in a restricted mode and leaves little margin for transient load events

From a marine power-distribution perspective, Raspberry Pi 5 must be treated as a **computer-class** lo device.

Translating 5 V Raspberry Pi Loads to NMEA 2000 Backbone Current

When a Raspberry Pi is powered from the NMEA 2000 backbone, the 12 V network power must be cor power supply (SMPS).

This conversion matters.

Approximate power relationships:

- Raspberry Pi 3:
 $5\text{ V} \times 2.5\text{ A} \approx 12.5\text{ W} \rightarrow$ roughly 1.1–1.3 A from the 12 V side
- Raspberry Pi 4:
 $5\text{ V} \times 3\text{ A} \approx 15\text{ W} \rightarrow$ roughly 1.4–1.6 A from the 12 V side
- Raspberry Pi 5:
 $5\text{ V} \times 5\text{ A} \approx 25\text{ W} \rightarrow$ roughly 2.3–2.7 A from the 12 V side

These values assume reasonable SMPS efficiency and do not include additional margin for transient lo

In a network commonly constrained to about 3 A total, a Raspberry Pi 5 alone can consume **most of th budget**.

PiCAN-M Variants and Power Behavior

The [PiCAN-M by Copperhill Technologies](#) is a compact and robust CAN bus interface designed specific used in marine and industrial applications. It provides a direct hardware connection to NMEA 2000 and enabling reliable reception and transmission of CAN frames for software platforms such as Signal K an seamlessly with the Raspberry Pi ecosystem, offering stable CAN communication, galvanic isolation op versions with or without an onboard 12 V to 5 V switch-mode power supply, allowing flexible integration architectures. Designed with real-world installations in mind, PiCAN-M is widely used as a dependable based systems and CAN-based marine networks.

The PiCAN-M is available in variants with and without an onboard SMPS.

PiCAN-M with onboard SMPS (NMEA-powered)

This variant includes a $12\text{ V} \rightarrow 5\text{ V}$ switch-mode power supply rated for 5 V at 3 A. The SMPS draws its NMEA 2000 network.

Behavior by Raspberry Pi version:

- Raspberry Pi 3
Generally suitable. The SMPS provides adequate current, and the resulting backbone load is usually moderately loaded networks.
- Raspberry Pi 4
Typically functional but operating at the limit. System stability depends heavily on backbone wiring and transient behavior. Marginal installations may exhibit occasional undervoltage symptoms.
- Raspberry Pi 5
Not suitable. The 5 V / 3 A limit is below the Pi 5's specified requirement, and the corresponding 12 V exceeds what many networks can reliably supply. Undervoltage warnings, reboots, or intermittent behavior are likely.

PiCAN-M without onboard SMPS (externally powered Raspberry Pi)

In this configuration, the Raspberry Pi is powered independently, and PiCAN-M interfaces only to the CAN bus.

Behavior by Raspberry Pi version:

- Raspberry Pi 3
Fully supported with a dedicated 5 V / 2.5 A supply or equivalent DC/DC converter.
- Raspberry Pi 4
Fully supported with a dedicated 5 V / 3 A supply.
- Raspberry Pi 5
Strongly recommended approach. A dedicated 5 V / 5 A-class supply provides the required headroom for the network from computer-class power demand.

This configuration aligns best with NMEA 2000 power-budget assumptions and results in the highest system reliability.

Typical Failure Patterns Caused by Marginal Power

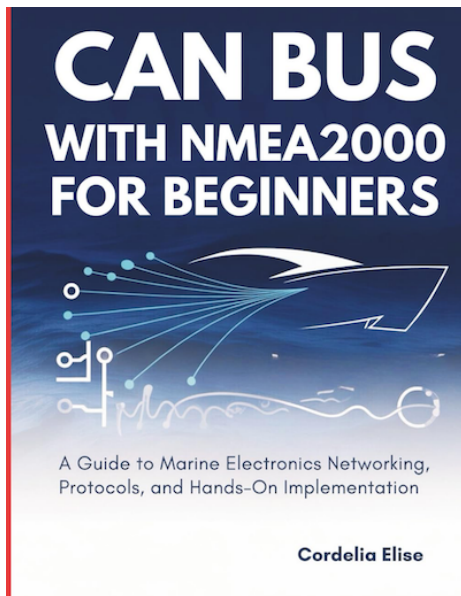
When Raspberry Pi power is undersized or drawn from an overstressed backbone, the symptoms are commonly:

- spontaneous reboots under load
- SD card filesystem corruption over time
- intermittent disappearance of NMEA 2000 devices
- unstable Wi-Fi or delayed startup
- undervoltage warnings logged by the operating system

These issues are frequently misattributed to software or CAN communication problems, when the root cause is power-related.

Practical Guidance for Marine Installations

- Raspberry Pi 3 and Raspberry Pi 4 can be used with PiCAN-M SMPS variants in suitable networks, well understood.
- Raspberry Pi 5 should be powered from a dedicated supply and not from NMEA 2000 backbone power.
- Treat NMEA 2000 power as a **network-device resource**, not as a general 5 V power rail.
- Separating computing loads from network power improves stability, simplifies troubleshooting, and eases installation practice.



CAN Bus with NMEA 2000 for Beginner Electronics Networking, Protocols, and Implementation

Modern marine electronics rely on NMEA 2000, a standard that allows engines, sensors, displays, navigation systems, and other components to communicate over a single, shared backbone. Instead of using multiple wires, it uses a robust two-wire CAN bus to distribute data efficiently. While powerful and flexible, this technology often appears daunting due to its terminology, strict wiring rules, and a lack of clear beginner-friendly resources.

Many boat owners, technicians, and engineers struggle to understand how NMEA 2000 systems actually work in practice. Confusing documentation, and overly theoretical explanations make it difficult to build a network with confidence. That is exactly why *First Steps in CAN Bus with NMEA 2000* was written. This book provides a beginner-friendly yet thorough understanding, designing, and maintaining NMEA 2000 networks without requiring prior CAN experience.

Whether you are new to marine electronics or looking to strengthen your foundation before moving on to more advanced topics, this guide walks you through every critical concept step by step. It explains how CAN communication works, how real devices share data on the network. Rather than acting as a dry reference manual, the book explains concepts through real-world examples, clear diagrams, and straightforward explanations.

Inside the book, you will learn how to design a stable network using correct cabling and termination practices, proper grounding to avoid common failures, understand PGNs and data flow, and safely integrate displays, sensors, and gateways. Dedicated chapters focus on troubleshooting and testing, showing how to diagnose voltage drops, network load problems using practical methods that reflect real installations.

Written with field-tested experience and a clear, professional voice, *First Steps in CAN Bus with NMEA 2000* turns the complex communication standard into something practical and approachable. It gives you the knowledge and confidence to maintain reliable NMEA 2000 networks, helping your systems perform as intended and saving time, frustration, and money the way. [More information...](#)

MARINE CAN BUS MARINE DATA SERVER NMEA 2000 GATEWAY NMEA 2000 POWER
OPENPLOTTER PISCAN-M RASPBERRY PI 5 POWER RASPBERRY PI MARINE APPLICATIONS
RASPBERRY PI NMEA 2000 SIGNAL K

Quick Links

- About Us
- Contact Us
- Shipping & Returns
- Press Room
- Privacy Policy
- Wiki
- Blog
- Documentation
- Controller Area Network (CAN) Development with the ESP32
- CAN, SAE J1939, NMEA 2000 Projects with the Arduino Due - Source Code Included
- PICAN CAN Bus HAT For Raspberry Pi - Selection Guide
- CANCrocodile - Contactless CAN Bus Monitoring
- Electronic Logging Device (ELD) Rule
- RSS Syndication

Categories

- Arduino
- Breakout Boards
- OBD-II & J1939 Cables
- CAN FD
- CAN Bus
- CANCrocodile
- Enclosures
- ESP32
- espBerry
- Gateways
- LIN Bus
- Literature
- NMEA 2000
- PiCAN HATs
- Raspberry Pi
- SAE J1939
- Software
- Technoton
- Teensy
- Special Deals
- Shipment

Brands

- SK Pang Electronics
- Copperhill Technologies
- Technoton
- WaveShare
- Yason
- Adafruit
- View all brands